

Core-resonant ionization of helium by intense XUV pulses: Analytical and numerical studies on channel-resolved spectral features

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Channel-resolved ionization of helium by laser pulses with a photon energy equal to the energy difference between the ionic ground- and first-excited states, i.e., in the core-resonant scenario, is investigated theoretically for a considered peak intensity of $\sim 10^{14}$ W/cm². By using the essential-states formalism, we derive analytical expressions for the channel-resolved photoelectron spectra, with the pulse-envelope effects taken into account. To examine the validity of the essential-states approach, we numerically solve the time-dependent Schrödinger equation of a model helium and use the corresponding results as references. The justified analytical estimation of the channel-resolved photoelectron spectra serves as a convenient approach to explore the effects of pulse durations and envelopes. It requires a longer pulse duration to observe the spectral peak splitting in the ionic first-excited-state channel than in the ionic ground-state channel. Ionization by two sequent pulses produces more complicated multipeak structures in the channel-resolved photoelectron spectra than ionization by a single pulse. Studies for the nonzero-detuning cases are also presented, where we propose a general procedure for understanding how the photoelectron spectra are built up.

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I. INTRODUCTION

Photoelectron spectroscopy of atoms, molecules, and solids has long been an important subject in the research of light-matter interactions [1]. It is a conventional tool to investigate the electronic structure of the targets. Also, photoelectron spectra (PES) can be used to explore the laser-induced dynamics, which could be ultrafast processes happening on femto- and attosecond timescales [2,3]. For a target with more than one electron, the PES may contain information of multielectron effects. In a two-electron system, for example, when two ionic states are resonantly coupled by one-photon transitions, a doublet would be observed in the spectra [4]. This effect, sometimes termed as core resonance [5–8], was first theoretically predicted by model calculations [4], and later experimentally observed for ionization of calcium by 400-nm lasers [9].

With the recent development of high-frequency light sources such as high-order harmonics and free electron lasers, one can produce intense laser pulses in the XUV and x-ray regimes [10–14]. This means that experimental research on the core-resonant ionization of fundamental two-electron systems, e.g., helium, is possible. For such a simple two-electron atom, one can focus on and isolate the basic mechanism of the core-resonance effect, since the system has much less complexity than a many-electron atom. As predicted in Ref. [15], the PES in the two resonantly coupled channels, i.e., the channel-resolved PES, may reveal different peak structures for specific laser-pulse parameters. Also, several theoretical studies have recently predicted that dynamic interference structures might be observed in the PES [16–21] when the ionization is induced by finite XUV pulses. Therefore it is meaningful to investigate how the channel-resolved PES of core-resonant ionization can be affected by pulse durations and envelopes.

Basically, there are two types of theoretical methods used in the previous studies on core-resonant ionization: (1) numerical simulations of the time-dependent Schrödinger equation (TDSE) on spatial grids, and (2) calculations performed within the essential-states formalism [22], where the time-dependent wave function is expanded in a limited basis of field-free eigenstates expected to capture the essential physics. In terms of the computational cost, the essential-states calculations are less demanding than the TDSE simulations. Also, the essential-states theory may allow one to obtain analytical results with some further approximations, providing useful insights into the dominant mechanisms in a given process. However, analytical calculations of the spectra based on the essential-states theory have been presented only for square pulses with a constant envelope (see, e.g., Refs. [4,6,7]). To consider nonconstant envelopes, researchers have so far resorted to numerical simulations (see, e.g., Refs. [7,15,20]). In contrast to the essential-states formalism, the TDSE approach is an *ab initio* method in which fewer simplifications are assumed. The TDSE calculations are therefore, in principle, more reliable and can be used as references to verify the validity of the approximations made in the essential-states theory [4,6]. Note that for the comparisons between the TDSE and the essential-states results in Refs. [4,6], the laser turn-on and turn-off effects have not been considered. Pulse-envelope effects related also to turn-on and turn-off are, however, quite important for the description of dynamic interference in the spectra. It is therefore desirable to develop analytical descriptions or simple estimations of the channel-resolved PES for core-resonant ionization with the pulse-envelope effects taken into account.

In this paper, we consider core-resonant ionization of helium (illustrated in Fig. 1) by finite XUV pulses and study how the channel-resolved PES are affected by pulse durations and envelopes. Our studies make use of both the essential-states

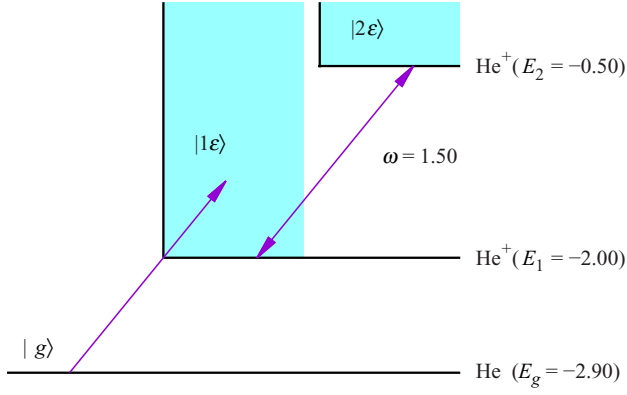


FIG. 1. Illustration for core-resonant ionization of He. As indicated by arrows, there are two dominant processes: (1) one-photon ionization from the ground-state He to the singly ionized continuum corresponding to the ground-state He^+ , and (2) resonant transitions between the two coupled continua.

and the TDSE approaches: first, we work in the essential-states framework to derive an analytical description of the channel-resolved PES for arbitrary pulse envelopes; then we examine the validity of the essential-states theory by comparing the analytical results with the TDSE results for a one-dimensional (1D) model of helium. Our comparisons demonstrate quantitative agreement. The validation of the analytical approach allows us to easily address the question of how channel-resolved PES are affected by pulse durations and envelopes. Although the analytical expressions are only available for the exact core resonance, we also investigate the nonzero-detuning cases with numerical calculations on the basis of the essential-states theory. Interestingly, the method used in treating the essential-states equations can provide a time-resolved perspective on how the channel-resolved PES are formed.

The paper is organized as follows. In Sec. II, we describe the considered processes and the theoretical methods used in this work. The results of the calculations are presented and discussed in Sec. III. Finally, we conclude with a brief summary. Atomic units (a.u.) are used throughout unless stated otherwise.

II. THEORY

A. Essential-states theory

To formulate the essential-states theory for the considered core-resonant ionization, we keep in mind the dominant processes shown in Fig. 1. The ground-state He can be ionized into the ground-state ion He^+ by absorbing one photon of energy $\omega = 1.50$, which resonantly couples two singly ionized continua corresponding to the ionic ground- and first-excited states. Following Ref. [4], we expand the time-dependent two-electron state as a linear combination of the essential states, i.e., the ground state $|g\rangle$ and two singly-ionized continua $|1\varepsilon\rangle$, $|2\varepsilon\rangle$:

$$|\Psi(t)\rangle = |g\rangle c_g(t) e^{-iE_g t} + \int d\varepsilon |1\varepsilon\rangle c_1(\varepsilon, t) e^{-i(E_1 + \varepsilon - \omega)t} + \int d\varepsilon |2\varepsilon\rangle c_2(\varepsilon, t) e^{-i(E_2 + \varepsilon - 2\omega)t}, \quad (1)$$

where E_g , E_j ($j = 1, 2$) and ε are energies of the ground state, the two ionic states and the free electron, respectively.

We consider a finite pulse over time $t \in [0, t_p]$, and assume that it is linearly polarized along the x axis. Within the dipole approximation, the pulse is described by the electric field

$$F(t) = F_0 f(t) \cos(\omega t), \quad (2)$$

where F_0 is the field strength and $f(t)$ is the envelope. In this work we restrict our discussion to many-cycle pulses and neglect the carrier-envelope-phase effects, since the core-resonance effect relies on the Rabi oscillations between the ionic states, the period of which is typically a few cycles. Within the rotating wave approximation (RWA), the effective Hamiltonian corresponding to the state in Eq. (1) reads

$$H_{\text{eff}}(t) = |g\rangle E_g \langle g| + \int d\varepsilon |1\varepsilon\rangle (E_1 + \varepsilon) \langle 1\varepsilon| + \int d\varepsilon |2\varepsilon\rangle (E_2 + \varepsilon) \langle 2\varepsilon| + \frac{F_0}{2} f(t) \left[e^{-i\omega t} \int d\varepsilon |1\varepsilon\rangle \langle 1\varepsilon| (x_1 + x_2) |g\rangle \langle g| + e^{-i\omega t} \iint d\varepsilon d\varepsilon' |2\varepsilon'\rangle \langle 2\varepsilon'| (x_1 + x_2) |1\varepsilon\rangle \langle 1\varepsilon| + \text{H.c.} \right], \quad (3)$$

where H.c. refers to the Hermitian conjugate.

For simplicity, we assume that the dipole couplings are real valued and positive, and introduce the notations

$$\langle 1\varepsilon | (x_1 + x_2) | g \rangle = D_g(\varepsilon), \quad (4)$$

$$\langle 1\varepsilon | (x_1 + x_2) | 2\varepsilon' \rangle \approx D_{12} \delta(\varepsilon - \varepsilon'). \quad (5)$$

Here the continuum-continuum coupling is approximated by a Dirac δ function, with the factor D_{12} being the dipole moment between the two ionic states. This description will be validated by comparisons with *ab initio* TDSE results.

The time evolution of the system is governed by the differential equations for the coefficients. Inserting Eq. (1) into the TDSE with the Hamiltonian [Eq. (3)] and projecting on the individual states, we obtain

$$i \frac{\partial}{\partial t} c_g(t) = \int d\varepsilon D_g(\varepsilon) \frac{F_0}{2} f(t) c_1(\varepsilon, t), \quad (6)$$

$$i \frac{\partial}{\partial t} c_1(\varepsilon, t) = (\varepsilon + E_1 - E_g - \omega) c_1(\varepsilon, t) + D_{12} \frac{F_0}{2} f(t) c_2(\varepsilon, t) + D_g(\varepsilon) \frac{F_0}{2} f(t) c_g(t), \quad (7)$$

$$i \frac{\partial}{\partial t} c_2(\varepsilon, t) = (\varepsilon + E_2 - E_g - 2\omega) c_2(\varepsilon, t) + D_{21} \frac{F_0}{2} f(t) c_1(\varepsilon, t), \quad (8)$$

with the initial condition $c_g(0) = 1$ and $c_j(\varepsilon, 0) = 0$ ($j = 1, 2$).

In the very special case of a constant envelope, one can easily solve Eqs. (6)–(8) by using Laplace transforms [4]. In general, however, the time dependence of $f(t)$ makes it much more complicated to seek for analytical solutions. Here our

approach to solve Eqs. (6)–(8) is inspired by Ref. [23]. We first make the “flat-continuum” assumption that D_g has a weak dependence on ε and approximate it by its value on resonance,

$$D_g \approx D_g(\varepsilon = \omega + E_g - E_1), \quad (9)$$

for the continuum-state energy range in which we are interested. By performing Fourier analysis with respect to the continuum-state energy

$$c_j(\varepsilon, t) = \int ds \exp(-i\varepsilon s) \kappa_j(s, t), \quad (j = 1, 2), \quad (10)$$

$$\kappa_j(s, t) = \frac{1}{2\pi} \int d\varepsilon \exp(+i\varepsilon s) c_j(\varepsilon, t), \quad (j = 1, 2), \quad (11)$$

we obtain the following equations:

$$i \frac{\partial}{\partial t} c_g(t) = \frac{D_g F_0}{2} f(t) 2\pi \kappa_1(s = 0, t), \quad (12)$$

$$\begin{aligned} i \frac{\partial}{\partial t} \kappa_1(s, t) = & -i \frac{\partial}{\partial s} \kappa_1(s, t) + (E_1 - E_g - \omega) \kappa_1(s, t) \\ & + \frac{D_{12} F_0}{2} f(t) \kappa_2(s, t) + \frac{D_g F_0}{2} f(t) c_g(t) \delta(s), \end{aligned} \quad (13)$$

$$\begin{aligned} i \frac{\partial}{\partial t} \kappa_2(s, t) = & -i \frac{\partial}{\partial s} \kappa_2(s, t) + (E_2 - E_g - 2\omega) \kappa_2(s, t) \\ & + \frac{D_{21} F_0}{2} f(t) \kappa_1(s, t). \end{aligned} \quad (14)$$

These equations can be solved by using the method of characteristics [23]. Let us introduce new variables $\eta = s$ and $\zeta = t - s$, and rewrite Eqs. (13) and (14) as

$$\begin{aligned} i \frac{d}{d\eta} \kappa_1(\eta, \zeta) = & (E_1 - E_g - \omega) \kappa_1(\eta, \zeta) \\ & + \frac{D_{12} F_0}{2} f(\eta + \zeta) \kappa_2(\eta, \zeta) \\ & + \frac{D_g F_0}{2} f(\eta + \zeta) c_g(\eta + \zeta) \delta(\eta), \end{aligned} \quad (15)$$

$$\begin{aligned} i \frac{d}{d\eta} \kappa_2(\eta, \zeta) = & (E_2 - E_g - 2\omega) \kappa_2(\eta, \zeta) \\ & + \frac{D_{21} F_0}{2} f(\eta + \zeta) \kappa_1(\eta, \zeta). \end{aligned} \quad (16)$$

In case of the exact resonance ($E_2 - E_1 = \omega$), one can easily decouple Eqs. (15) and (16) by introducing $\kappa_{\pm} = (\kappa_1 \pm \kappa_2)/\sqrt{2}$ and then find the analytical solutions by integrating with respect to η . The Dirac δ function in Eq. (15), which is due to the “flat-continuum” assumption, greatly simplifies the results. With the Rabi frequency denoted by $\Omega_0 = D_{12} F_0$, the solutions to Eqs. (13) and (14) are

$$\begin{aligned} \kappa_1(s, t) = & -i \frac{D_g F_0}{2} [\theta(s) - \theta(s - t)] f(t - s) c_g(t - s) \\ & \times \left\{ \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(-\varepsilon_0 + \frac{\Omega_0}{2} f(s') \right) \right] \right. \\ & \left. + \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(-\varepsilon_0 - \frac{\Omega_0}{2} f(s') \right) \right] \right\}, \end{aligned} \quad (17)$$

$$\begin{aligned} \kappa_2(s, t) = & -i \frac{D_g F_0}{2} [\theta(s) - \theta(s - t)] f(t - s) c_g(t - s) \\ & \times \left\{ \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(-\varepsilon_0 + \frac{\Omega_0}{2} f(s') \right) \right] \right. \\ & \left. - \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(-\varepsilon_0 - \frac{\Omega_0}{2} f(s') \right) \right] \right\}, \end{aligned} \quad (18)$$

where $\varepsilon_0 = E_g - E_1 + \omega = E_g - E_2 + 2\omega$, and Heaviside θ functions are introduced to restrict the variable range.

With $\kappa_1(s = 0, t) = -i \frac{D_g F_0}{4} f(t) c_g(t)$ substituted into Eq. (12), one obtains the solution to Eq. (6):

$$c_g(t) = \exp \left[-\frac{\pi D_g^2 F_0^2}{4} \int_0^t dt' f^2(t') \right]. \quad (19)$$

The solutions to Eqs. (7) and (8) are determined by Fourier transforms [Eq. (10)]:

$$\begin{aligned} c_1(\varepsilon, t) = & -i \frac{D_g F_0}{2} \int_0^t ds f(t - s) c_g(t - s) \\ & \times \left\{ \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(\varepsilon - \varepsilon_0 + \frac{\Omega_0}{2} f(s') \right) \right] \right. \\ & \left. + \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(\varepsilon - \varepsilon_0 - \frac{\Omega_0}{2} f(s') \right) \right] \right\}, \end{aligned} \quad (20)$$

$$\begin{aligned} c_2(\varepsilon, t) = & -i \frac{D_g F_0}{2} \int_0^t ds f(t - s) c_g(t - s) \\ & \times \left\{ \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(\varepsilon - \varepsilon_0 + \frac{\Omega_0}{2} f(s') \right) \right] \right. \\ & \left. - \frac{1}{2} \exp \left[-i \int_{t-s}^t ds' \left(\varepsilon - \varepsilon_0 - \frac{\Omega_0}{2} f(s') \right) \right] \right\}. \end{aligned} \quad (21)$$

The channel-resolved PES after the pulse ($t \geq t_p$) are

$$\frac{dP_j}{d\varepsilon} = |c_j(\varepsilon, t_p)|^2, \quad (j = 1, 2). \quad (22)$$

We will compare the channel-resolved PES calculated by Eq. (22) with the *ab initio* TDSE results in Sec. III.

B. TDSE and numerical methods

The TDSE simulations are performed for the widely used 1D two-electron model [24–26] with the Hamiltonian

$$H(t) = \sum_{j=1}^2 \left\{ \frac{1}{2} \left[-i \frac{\partial}{\partial x_j} + A(t) \right]^2 + V_{\text{en}}(x_j) \right\} + V_{\text{ee}}(x_1, x_2). \quad (23)$$

Here

$$V_{\text{en}}(x_j) = \frac{-Z_{\text{en}}}{\sqrt{x_j^2 + a}}, \quad (j = 1, 2), \quad (24)$$

$$V_{\text{ee}}(x_1, x_2) = \frac{Z_{\text{ee}}}{\sqrt{(x_1 - x_2)^2 + a}}, \quad (25)$$

are the electron-nucleus and the electron-electron interactions, which are softened to avoid the Coulomb singularity. Such a 1D model allows us to study two-electron effects with much less computational difficulty than the three-dimensional (3D) simulations. For core-resonant ionization by linearly polarized laser pulses, the 1D *ab initio* calculations can be used as practical references to examine the accuracy of the essential-states theory. By choosing parameters $Z_{\text{en}} = 1.1225$, $Z_{\text{ee}} = 0.6317$, and $a = 0.09169$, we obtain the He ground-state energy $E_g = -2.90$, He⁺ ground-state energy $E_1 = -2.00$, and first-excited-state energy $E_2 = -0.50$, which are in good agreement with the values for the real system. The same parameters were also used in Ref. [27] for simulating double ionization of helium.

The laser interaction in Eq. (23) is described in velocity gauge by the vector potential

$$A(t) = - \int_{-\infty}^t dt' F(t'), \quad (26)$$

which fulfills $A(-\infty) = A(+\infty) = 0$ [28,29]. In our calculations, the electric field $F(t)$ in Eq. (2) has a sine-squared envelope

$$f(t) = \sin^2 \left(\frac{\pi t}{t_p} \right) [\theta(t) - \theta(t - t_p)], \quad (t_p = 2\pi N_c / \omega), \quad (27)$$

with an integer number of cycles N_c .

For the TDSE simulations, we discretize the simulation volume by employing a finite-element discrete variable representation (FEDVR) [30–32] combined with infinite-range exterior complex scaling (irECS) [33]. The coordinate space is divided into inner and outer regions by a complex scaling radius r_c , which is chosen to be 50 for the calculations in this work. The inner region $[-r_c, r_c]$ is discretized by finite elements with a fixed equal size of 2 and 11 Gauss-Lobatto basis functions in each element. The outer regions $(-\infty, -r_c]$ and $[r_c, +\infty)$ are complex scaled with a scaling angle $\theta = 0.3$ and discretized by 30 basis functions. (See Refs. [33–35] for details of the irECS method and the basis functions.)

We propagate the TDSE with the Arnoldi-Lanczos time-propagator [36,37] with a time step of $\Delta t = 0.002$. The initial state is obtained by imaginary time propagation [38]. The channel-resolved PES are extracted by using the time-dependent surface flux (tSURFF) method [39,40] with the tSURFF surfaces placed at $x = \pm 45$. We emphasize that the irECS method works as an efficient absorber so that the wave function in the inner region remains accurate for applying the tSURFF method. The use of velocity gauge in the TDSE calculations is required by the irECS method [33]. Note that recently, this methodology was used for investigating high-order above-threshold ionization of helium in the long-wavelength regime [41].

III. RESULTS

A. Essential states vs TDSE

Let us begin our discussion with a comparison between some essential-states and TDSE results. The TDSE calculations are performed for a fixed field strength $F_0 = 0.0534$,

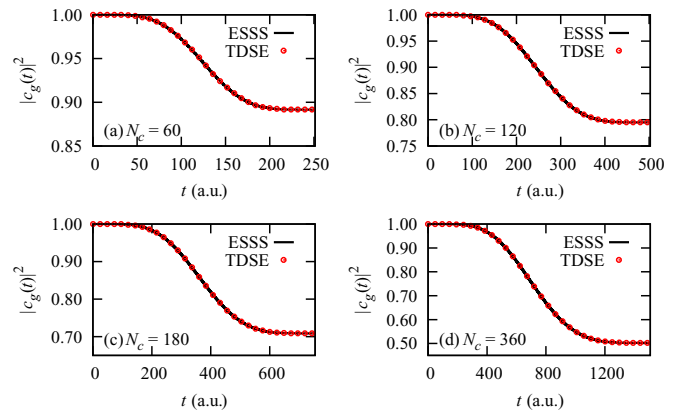


FIG. 2. Time-dependent ground-state probability obtained by the essential-states (labeled as ESSS) and the TDSE calculations for a fixed laser frequency $\omega = 1.50$ and different pulse durations.

which corresponds to a peak intensity of 10^{14} W/cm². Such laser pulses can yield a large amount of single ionization with a near-zero probability for double ionization ($< 10^{-4}$), for which we expect the essential-states expansion [Eq. (1)] to be valid. The dipole moment between the two ionic states for the 1D model is $D_{12} = 0.4823$. To obtain the essential-states results by using Eqs. (19)–(21), we also need to know the value of D_g . This value can be extracted by fitting the norm square of Eq. (19) at the end of the pulse to the final ground-state probability obtained from the corresponding TDSE calculation. We find $D_g = 0.5213$ for $\omega = 1.50$ from a single calculation with $N_c = 60$ and then apply this value to other essential-states calculations with different N_c .

Figure 2 shows the time-dependent ground-state probability obtained from Eq. (19) and that obtained by projection on the ground state in the TDSE calculations. We find excellent agreement between the analytical (essential-states) and the numerical (TDSE) results for different N_c , which proves that the essential-states theory can indeed describe the time evolution of the ground-state population for different pulse durations. Note that the drop of the ground-state probability can reach ~ 0.5 for a relatively long pulse with $N_c = 360$. In this sense we say that the considered laser pulses are intense, although they are not extremely strong to cause considerable double ionization. Since our aim here is mainly to examine the analytical approach for pulsed laser interaction, we do not consider extremely high intensity, e.g., $\sim 10^{16}$ W/cm², which would invalidate the present essential-states expansion [Eq. (1)] not accounting for double ionization.

Comparisons of the channel-resolved PES between the essential-state and the TDSE calculations are presented in Fig. 3. For the selected pulse durations, we display four types of representative spectral features: (a) only a single peak is observed in both channels; (b) two peaks are observed in channel 1 (the ionic ground-state channel) while only one peak is observed in channel 2 (the ionic first-excited-state channel); (c) two peaks are observed in both channels; and (d) the spectra manifest multipoint structures, i.e., pronounced dynamic interference features. In all these cases, the essential-states and the TDSE results show quantitative agreement, which justifies the analytical estimation of the channel-resolved PES

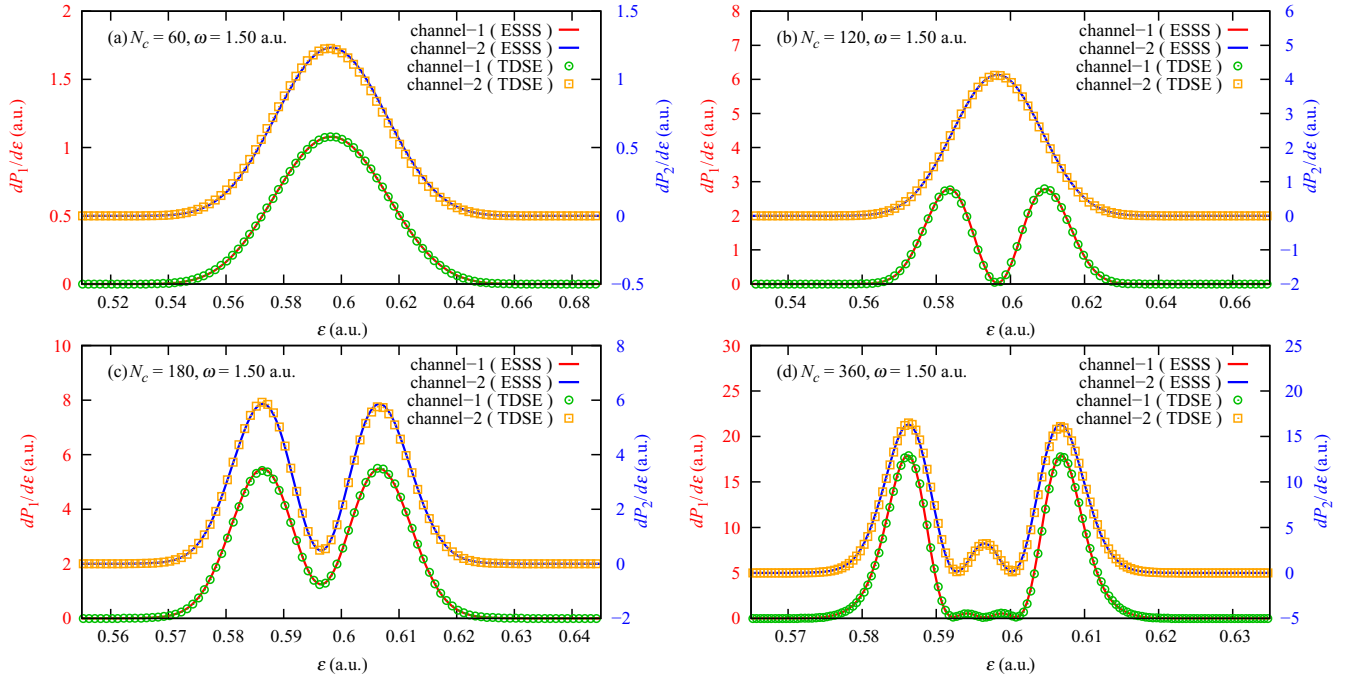


FIG. 3. Channel-resolved PES obtained by the essential-states (labeled as ESSS) and the TDSE calculations for a fixed frequency $\omega = 1.50$. Channel 1 (channel 2) denotes ionization into the ionic ground-state (first-excited-state) channel and its vertical axis is to the left (right). The four selected pulse durations are the same as in Fig. 2.

based on Eqs. (20) and (21). The different types of spectral features in Fig. 3 imply that for the considered pulses it might be more appropriate to view the spectral features as dynamic interference [20] rather than only peak splitting.

B. Effects of pulse durations and envelopes

The justified analytical estimation of the channel-resolved PES can offer a convenient approach to explore the effects of pulse durations and envelopes. A doublet in the PES, which originates from the Rabi splitting of two ionic states, has been considered as a key feature of core-resonant ionization [4]. For the effect of pulse durations on the channel-resolved PES, it is natural to ask what the minimum pulse duration is for observing a doublet in each channel. One necessary condition is that the spectral width should be smaller than the doublet splitting, where the latter equals the Rabi frequency $\Omega_0 = D_{12}F_0$ [4]. For the considered sine-squared envelope [Eq. (27)], the spectral width (defined as the full width at half maximum) is approximately $1.44\omega/N_c$ [42]. Therefore a minimum number of cycles $N_c \approx 84$ is required for $F_0 = 0.0534$. In fact, this condition determines only the minimum pulse duration for observing a doublet in channel 1, while more cycles are required for observing a doublet in channel 2, a point we will come back to below.

To study how the spectral features are affected by the pulse durations, we apply the analytical estimation of the channel-resolved PES for a series of N_c ranging from 40 to 400 and display the results in Fig. 4. The spectral intensity in this plot is divided by the corresponding total ionization probability $[1 - |c_g(t_p)|^2]$, which means that the total integrated spectral intensity (channel 1 plus channel 2) is

normalized to 1. We find that the PES in both channels evolve from single-peak to multipeak features as the pulse duration increases. A minimum number of cycles $N_c \approx 84$ is needed to identify two peaks in the PES of channel 1, which is consistent with the above condition determined by the spectral width. Comparatively, a minimum number of cycles $N_c \approx 144$ is needed to identify two peaks in the PES of channel 2. Since the essential-state theory indicates that ionization initially happens in channel 1, it would take more time for the free electron (and also the spectral features) to appear in channel 2 than in channel 1.

In Ref. [15] the number of peaks in the spectra was related to the number of Rabi cycles, or the pulse area $\Omega_0 \int dt f(t)$. Here we would like to emphasize that linking the number of peaks in the spectra to the pulse area is reasonable only for a single pulse with one pair of rising and falling edges. For more complicated envelopes with more than one pair of rising and falling edges, the dynamic interference features in the spectra can hardly be related to the pulse area. To demonstrate this and to model a pump-probe scenario, we show in Fig. 5 the channel-resolved spectra for two sequent pulses, i.e., a combined envelope of the form $f_2(t) = f(t) + f(t - t_p)$ with $f(t)$ given by Eq. (27). The two pulses used in Fig. 5(a) are identical to the single pulse used in Fig. 3(a), and the total pulse area for Fig. 5(a) is the same as that for Fig. 3(b). The multipeak structures observed in Fig. 5(a), however, resemble neither Fig. 3(a) nor Fig. 3(b). Therefore the pulse area is not very helpful for predicting the number of spectral peaks in this case. The same argument applies when one compares Fig. 5(b) with Figs. 3(c) and 3(d). Therefore in this work we just briefly conclude that it requires a longer pulse to observe multipeak spectral features in the ionic first-excited-state channel than

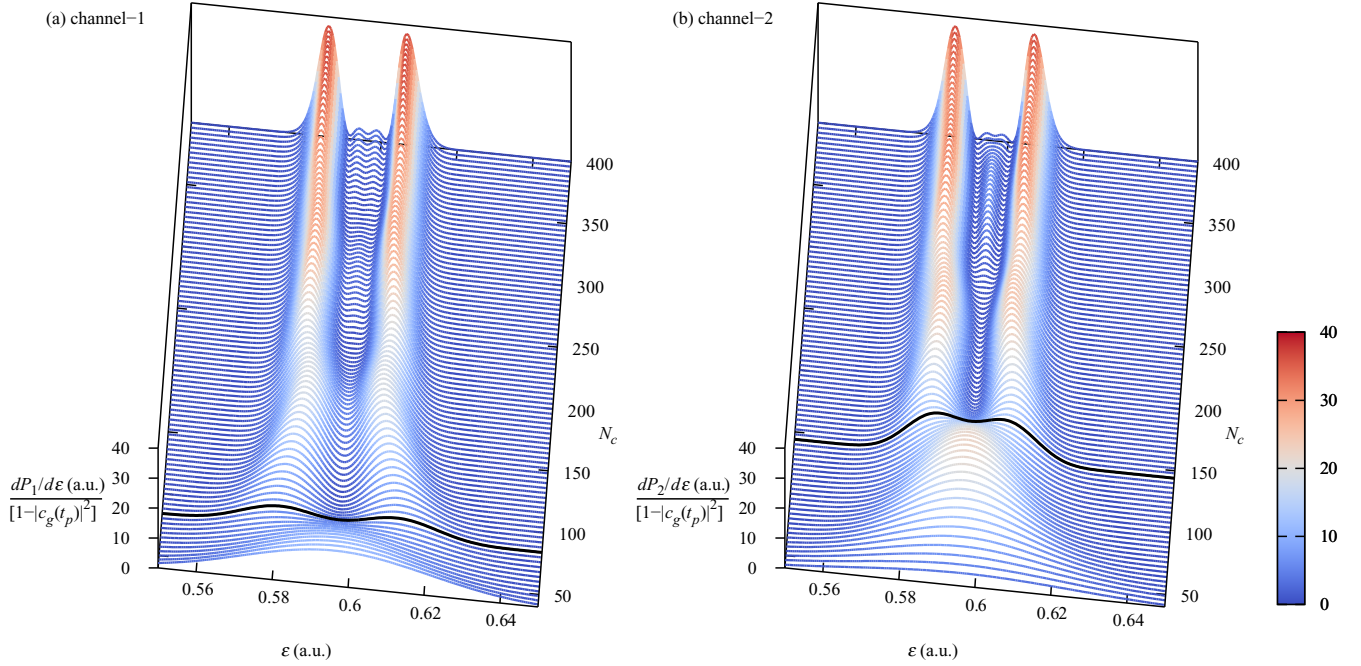


FIG. 4. Normalized channel-resolved PES for a series of different number of cycles (from 40 to 400), obtained by the essential-states calculations for a fixed laser frequency $\omega = 1.50$. Channel 1 (channel 2) denotes ionization into the ionic ground-state (first-excited-state) channel. The solid black curves indicate the minimum pulse duration for which one can visually observe two peaks in the spectra.

in the ionic ground-state channel, without making further analysis based on the pulse area. The difference in the pulse duration needed can be understood by noting that ionization into the ionic first-excited-state channel only proceeds after

ionization into the ionic ground-state channel, and it takes some time for population to build up.

Again, the essential-states and the TDSE results in Fig. 5 manifest excellent agreement. This means that the analytical estimation of the channel-resolved PES has a wide applicability for various and complicated pulse envelopes.

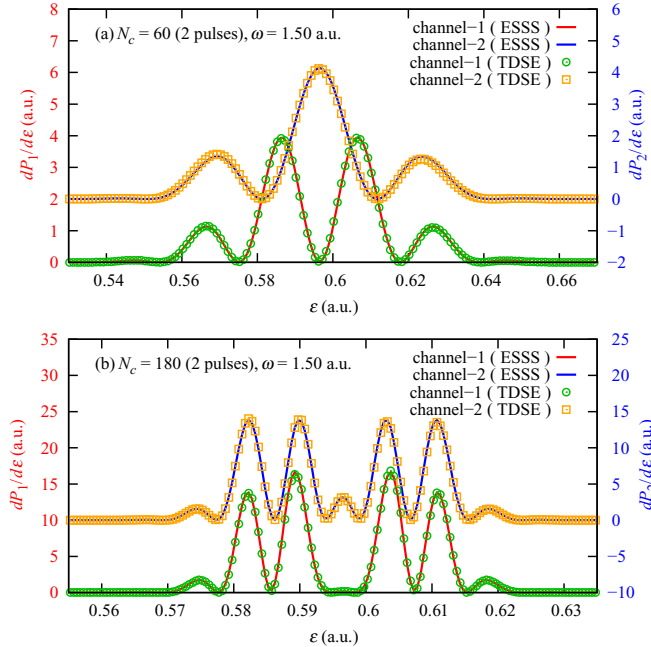


FIG. 5. Channel-resolved PES obtained by the essential-states (labeled as ESSS) and the TDSE calculations for two sequent pulses with frequency $\omega = 1.50$. Channel 1 (channel 2) denotes ionization into the ionic ground-state (first-excited-state) channel and its vertical axis is to the left (right).

C. Studies for the nonzero-detuning cases

The analytical expressions presented in Eqs. (20) and (21) are only valid for the exact core resonance. In case of a small but nonzero detuning $\Delta = E_2 - E_1 - \omega$, Eqs. (15) and (16) cannot be decoupled, and in this case we do not have analytical solutions. The method of characteristics used in Sec. II A, however, still allows us to determine the channel-resolved PES by numerical calculations. In particular, as we will now discuss, this method can offer a time-resolved perspective on how the PES are built up.

To evaluate the spectra at the end of the pulse t_p , we solve Eqs. (15) and (16) for a series of $\kappa_j(\eta = s, \zeta = t_p - s)$ with the corresponding initial condition $\kappa_j(\eta = s - t_p, \zeta = t_p - s) = 0$ ($j = 1, 2$). Here the variables η and ζ are introduced in the same way as in Sec. II A, and s is any time within the pulse duration ($0 < s < t_p$). Note that the source term in Eqs. (15) and (16) is a Dirac δ function, so we can in practice use the initial condition

$$\begin{bmatrix} \kappa_1(\eta = 0_+, \zeta = t_p - s) \\ \kappa_2(\eta = 0_+, \zeta = t_p - s) \end{bmatrix} = \begin{bmatrix} -i \frac{D_g F_0}{2} f(t_p - s) c_g(t_p - s) \\ 0 \end{bmatrix}, \quad (28)$$

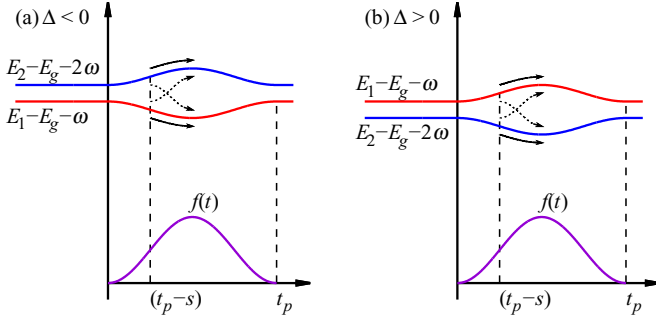


FIG. 6. Dynamics of the coupled continua for the nonzero-detuning cases, illustrated with the field-dressed energy levels of the Hamiltonian $\underline{\mathbf{H}}$ given by Eq. (30). The time-dependent eigenenergies of $\underline{\mathbf{H}}$ are displayed by red and blue curves, which respectively correspond to channel 1 (the ground-state He^+) and channel 2 (the first-excited state He^+) when the pulse is off. In this field-dressed picture, the free electron born at time $(t_p - s)$ will undergoes adiabatic and nonadiabatic processes indicated by solid and dotted lines with arrows, respectively.

and propagate the coupled homogeneous equations (for $\eta > 0$)

$$i \frac{d}{d\eta} \begin{bmatrix} \kappa_1(\eta, \zeta = t_p - s) \\ \kappa_2(\eta, \zeta = t_p - s) \end{bmatrix} = \underline{\mathbf{H}}(t_p - s + \eta) \begin{bmatrix} \kappa_1(\eta, \zeta = t_p - s) \\ \kappa_2(\eta, \zeta = t_p - s) \end{bmatrix} \quad (29)$$

with the 2×2 Hamiltonian matrix $\underline{\mathbf{H}}$ given by

$$\underline{\mathbf{H}}(t) = \begin{bmatrix} E_1 - E_g - \omega & \Omega_0 f(t)/2 \\ \Omega_0 f(t)/2 & E_2 - E_g - 2\omega \end{bmatrix}. \quad (30)$$

Note that Eq. (19) is still valid, as one can directly find $\kappa_1(s = 0, t) = [\kappa_1(0_-, t) + \kappa_1(0_+, t)]/2$ by integrating Eq. (15) with respect to η over an infinitesimal range $[0_-, 0_+]$.

One can understand the process in the following way. At time $(t_p - s)$, the system makes a transition (“jump”) from the ground state $|g\rangle$ to the first continuum $|1\varepsilon\rangle$ with amplitude

$$\begin{bmatrix} \kappa_1^A(\eta = \eta', \zeta = t_p - s) \\ \kappa_2^A(\eta = \eta', \zeta = t_p - s) \end{bmatrix} = \underline{\mathbf{U}}(t_p - s + \eta') \exp \left\{ -i \int_{t_p - s}^{t_p - s + \eta'} dt' \begin{bmatrix} \lambda_1(t') & 0 \\ 0 & \lambda_2(t') \end{bmatrix} \right\} \underline{\mathbf{U}}^\dagger(t_p - s) \begin{bmatrix} \kappa_1(\eta = 0_+, \zeta = t_p - s) \\ \kappa_2(\eta = 0_+, \zeta = t_p - s) \end{bmatrix}, \quad (32)$$

where the superscript A on the amplitudes κ_j^A ($j = 1, 2$) denotes adiabatic evolution in the field-dressed picture.

We apply our presented essential-states procedure for two selected laser frequencies $\omega = 1.49$ and $\omega = 1.51$ with a fixed number of cycles $N_c = 180$, and compare the channel-resolved PES with the corresponding TDSE results in Fig. 7. Compared with the exact core-resonant cases (Figs. 3 and 5), we see that the nonzero detuning breaks the symmetry of the spectral peaks. One can also find in Fig. 7 that the PES in channel 1 has higher spectral intensity than that in channel 2, as the nonzero detuning attenuates the amplitude of the Rabi oscillations. We also show in Fig. 7 an additional group of

$-i \frac{D_g F_0}{2} f(t_p - s) c_g(t_p - s)$. Then this part of the photoelectron is governed by the time-dependent Hamiltonian of the coupled continua [Eq. (30)], undergoing Rabi oscillations between the two ionic channels. All such contributions are coherently summed up according to Eq. (10), with the factor $\exp(-i\varepsilon s)$ accounting for the accumulated phase of the free electron from time $(t_p - s)$ to t_p . With $c_j(\varepsilon, t_p)$ ($j = 1, 2$) numerically evaluated, one can then use Eq. (22) to calculate the channel-resolved PES at the end of the pulse t_p . The time-resolved picture described here also includes the exact core resonance as a special case where the unitary transformation for diagonalizing $\underline{\mathbf{H}}$ is time independent, leading to an analytically solvable Rabi model. Although analytical expressions are not generally available, the presented formalism clearly shows that the coherent sum of all contributions during the pulse is the origin of dynamic interference in the spectra.

As shown in Fig. 6, we can also illustrate the process with the time-dependent eigenenergies of $\underline{\mathbf{H}}$, which reflect the field-induced energy shifts. The field-free and field-dressed eigenstates are connected by the unitary transformation for diagonalizing $\underline{\mathbf{H}}$:

$$\underline{\mathbf{U}}^\dagger(t) \underline{\mathbf{H}}(t) \underline{\mathbf{U}}(t) = \begin{bmatrix} \lambda_1(t) & 0 \\ 0 & \lambda_2(t) \end{bmatrix}. \quad (31)$$

Thus the “jump” into the field-free continuum $|1\varepsilon\rangle$ at time $(t_p - s)$ can be expressed in terms of the field-dressed continua by applying $\underline{\mathbf{U}}^\dagger(t_p - s)$ to Eq. (28). For a small but nonzero detuning Δ , the free electron born at time $(t_p - s)$ will not completely follow the field-dressed continua adiabatically; nonadiabatic processes will also occur due to the time dependence of the unitary transformation [Eq. (31)]. In pursuit of accurate estimations of the channel-resolved PES, we emphasize, and this will be demonstrated by an example below, that it is necessary to consider both adiabatic and nonadiabatic dynamics of the field-dressed continua when numerically solving Eq. (29). Mathematically, this means that it is inexact to assume purely adiabatic dynamics corresponding to the following equation:

essential-states results that are obtained by neglecting nonadiabatic processes in the field-dressed picture using Eq. (32). Within this adiabatic approximation, the estimated spectra do not deviate too much from the reference results. Some channel-resolved spectral features are, however, not well described when assuming purely adiabatic dynamics of the field-dressed continua. With nonadiabatic processes neglected, the primary spectral peak in channel 1 is overestimated while the secondary spectral peak in channel 2 is missing.

The excellent agreement between the essential-states and the TDSE results observed in Fig. 7 validates our procedure for calculating the channel-resolved PES. Key to our

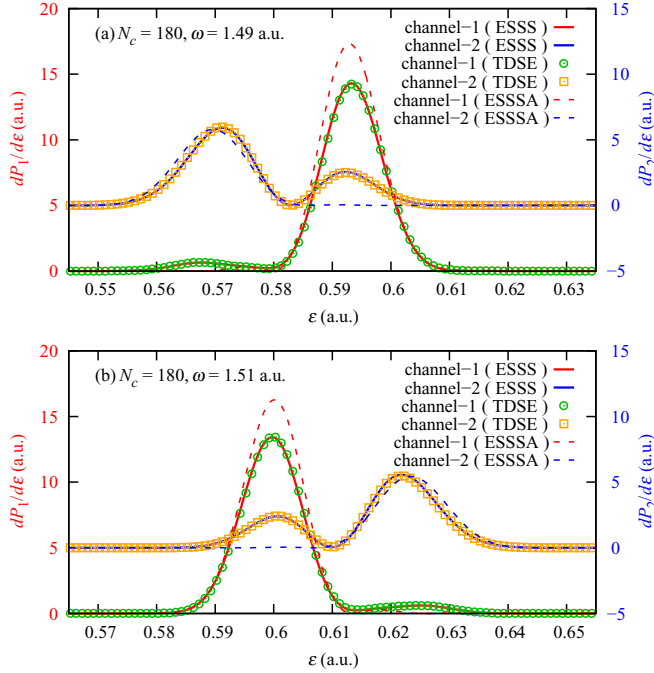


FIG. 7. Channel-resolved PES obtained by the essential-states (labeled as ESS) and the TDSE calculations for slightly different laser frequencies (a) $\omega = 1.49$ and (b) $\omega = 1.51$. Channel 1 (channel 2) denotes ionization into the ionic ground-state (first-excited-state) channel and its vertical axis is to the left (right). Dashed curves (labeled as ESSA) are obtained by neglecting nonadiabatic processes of the field-dressed continua in the essential-states calculations.

procedure is that the system is first ionized into channel 1 (the field-free continuum $|1\varepsilon\rangle$) and then the photoelectron can be coherently redistributed in different channels by the time-dependent Hamiltonian of the ionic system. This idea can also be extended to cases with more than two continua involved (see, e.g., the essential-states expansion used in Ref. [15]). We mention that a different simple estimation of the PES was proposed in Ref. [20]. Comparatively, our approach is more accurate. (See Fig. 6 in Ref. [20], where the disagreement between the estimated and computed spectra can be clearly observed.)

In closing this section, we note that the essential-states approach can also be extended to 3D cases, for which one should use the corresponding values for the dipole couplings. The validation of the essential-states approach shown in this work indicates that it could be applicable for modeling core-resonant ionization of real helium atoms by laser pulses

with a peak intensity of $\sim 10^{14}$ W/cm². In addition to easy interpretation of the physical processes, the essential-state approach also offers a significant reduction in the computational cost for evaluation of the spectra. For example, it takes only ~ 2 minutes to obtain the essential-state results shown in Fig. 7 on a single CPU core, while the corresponding TDSE calculations require a few hours for multicore processors. For simulating 3D cases, this speedup would be even more favorable. Although it is difficult to justify the essential-state approach by *ab initio* calculations for larger systems, the present essential-state approach would be generally applicable when the following conditions are fulfilled: (1) the ionic ground state and an isolated ionic excited state are resonantly coupled by a photon energy that can simultaneously ionize the neutral by single-photon absorption; and (2) the laser intensity causes no considerable multiple ionization, i.e., the intensity is not extremely high.

IV. CONCLUSION

We studied core-resonant ionization of helium by laser pulses with a peak intensity of 10^{14} W/cm². With the pulse-envelope effects taken into account, we derived analytical expressions for the channel-resolved PES on the basis of the essential-states theory. We justified the validity of the essential-states approach by comparing the results with those obtained from *ab initio* calculations performed for a 1D model helium. Therefore the analytical estimation of the channel-resolved PES can be applied to explore the effects of pulse durations and envelopes. For example, a longer pulse duration is required for observing the spectral peak splitting in the ionic first-excited-state channel than in the ionic ground-state channel. It was also found that ionization by two sequent pulses produces more complicated multipeak structures in the channel-resolved PES than ionization by a single pulse. For the nonzero-detuning cases where no analytical expressions are available, the essential-state approach still allows us to numerically calculate the channel-resolved PES at low computational cost. Importantly, the method of characteristics used in treating the essential-states equations provides a time-resolved perspective on how the spectra are formed.

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